

A Tree Code for Quantum Error Correction — roadmap handout

Rowan Brad Quni-Gudzinis — QNFO · CWI Summer School poster session, Wed 26 Aug 2026 · full plan: docs/tree-code-roadmap.md in github.com/QNFO/cwi-qec-poster-2026

The claim

Holographic perfect-tensor codes on a truncated Bruhat–Tits tree. Computed thresholds (analytical recursion + exact stabilizer simulation + Monte Carlo, re-verified by a deposited 35-check script): bit-flip **50.0%** ($4.6\times$ surface), depolarizing **75.0%** ($75\times$), independent $X+Z$ **17.30%** ($0.92\times$ vs toric $\sim 18.9\%$). **No hardware demonstration yet.** Bit-flip and $X+Z$ are i.i.d. code-capacity; the 75% is the code's structured-noise simulation (channel pending — D3).

Where the evidence stands

- Verified computationally, three independent methods; the 35-check verification script is deposited with the guide record.

- Boundary: under independent errors the qudit family $\approx 2 \times 10^{-4}$ ($\sim 55\times$ below surface); the claim targets correlated-failure regimes — test T1 probes exactly this.
- An external no-go was engaged: QEC–Darwinism tradeoff audited on tree spaces; boundary moves to $F_L \approx 0.83\text{--}0.85$, a residual gap remains.
- Self-correction: a naive p-adic valuation reading of stabilizer codes carries no new content.
- No hardware yet — everything above is classical simulation.

What we test next

#	Test	Where	Disconfirmation criterion
T1	Matched-noise threshold comparison vs surface	Classical, days–weeks	Advantage falls to parity under matched noise
T2	Decoder complexity + full overhead	Classical	Advantage disappears with decoding cost
T3	Random-walk return probability on p-regular trees	Analog/ion, ~ 8 wk	No change in decay exponent
T4	Clock-rest ultrametricity split	Trapped ion, ~ 8 wk	Split outside predicted range

(0% vs 29–35%)

T5 Independent re-
implementation
of T1

Community

Reproducibility
test

What we'd like from you

1. An openly specified **correlated-noise benchmark** (model + data) for T1 · 2. decoder-complexity reality check for tree-structured codes · 3. pointers on **perfect tensors for $p > 2$** · 4. an **independent re-run of T1** — the simulator and protocol are handed over.

DOIs (verified 21 Aug 2026): 10.5281/zenodo.20109836 (architecture) · 10.5281/zenodo.22038733 (guide + verification script) · 10.5281/zenodo.21046993 (qudit QEC) · 10.5281/zenodo.22025544 (trapped-ion testbed) · 10.5281/zenodo.21809888 (QEC–Darwinism audit) · 10.5281/zenodo.21979060 (prime-valuation correction)